

A PROJECT TITLE

ON

Credential Stuffing Attack Detection Using GBM and Isolation Forest

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

Name: HRUDAY DEEPAK

Roll: 2520030138

Name: B . SURYA PRAKASH REDDY

Roll: 2520030145

Name: K . AASRITHA

Roll: 2520030178

*Under the Esteemed guidance of
Mr. Gaddam Ravindra Babu
Assistant Professor*

CSE



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043

Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

Name of Title: Credential Stuffing Attack Detection Using GBM and Isolation Forest

1. ABSTRACT:

Cybersecurity is one of the most critical concerns for online platforms as it protects user accounts and sensitive data from unauthorized access. Credential stuffing is a major security threat in which attackers use stolen username-password pairs obtained from previous data breaches to gain unauthorized access to multiple online accounts through automated login attempts. Detecting such attacks accurately helps organizations prevent account takeovers, protect user privacy, and maintain trust in their platforms. Machine Learning provides a reliable way to analyze login activity data and detect credential stuffing attacks with better accuracy. This project, "**Credential Stuffing Attack Detection Using GBM and Isolation Forest**," aims to develop a machine learning-based system for identifying suspicious login attempts using login and network related features. The dataset is first cleaned and prepared by handling missing values and selecting the required features such as login frequency, IP reputation, request rate, and failed login patterns. Then, two algorithms—Gradient Boosting Machine (GBM) for supervised classification of malicious login attempts and Isolation Forest for unsupervised anomaly detection—are used to train and test the detection models. The performance of these models is compared using evaluation metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC to identify the model that gives the best detection accuracy. The main goal of this project is to build a reliable and efficient credential stuffing detection system that can help organizations safeguard user accounts in real time. This project also demonstrates how Machine Learning can be applied in cybersecurity to strengthen authentication systems, reduce fraud, and support safer digital platforms.

2. INTRODUCTION:

- Cybersecurity is one of the most important concerns in the digital economy as online platforms store and manage sensitive user information, financial data, and personal credentials. It plays a vital role in ensuring user trust and protecting digital infrastructure. However, login systems are frequently targeted by attackers using automated tools, bots, and leaked credential databases, making it difficult for organizations to distinguish legitimate users from malicious login attempts, which may result in account takeovers, data breaches, and financial losses.
- With the advancement of technology, Machine Learning has become a powerful tool for solving problems in cybersecurity by analyzing large volumes of login traffic and identifying suspicious behavioral patterns. Machine learning algorithms can identify hidden patterns and relationships between login attributes such as IP address, device fingerprint, login frequency, and time intervals, helping security teams make informed decisions regarding access control, rate limiting, and account protection.

- In this project, "Credential Stuffing Attack Detection Using GBM and Isolation Forest," machine learning techniques are used to detect credential stuffing attacks based on login and network related data. The dataset is preprocessed by handling missing values and selecting relevant features before training the detection models. Two algorithms—Gradient Boosting Machine (GBM) and Isolation Forest—are implemented and compared to determine which model provides the highest detection accuracy. The performance of these models is evaluated using metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC. The main objective of this project is to develop a reliable and accurate credential stuffing detection system that helps organizations strengthen authentication security, reduce fraudulent login attempts, and promote safer and more resilient digital platforms.

3. SOFTWARE REQUIREMENTS:

The following software is used to develop this project:

- Operating System: Windows 10/11 or Ubuntu Linux
- Programming Language: Python 3
- Development Tool: Jupyter Notebook, Google Colab, or Visual Studio Code (VS Code)
- Machine Learning Library: Scikit-learn
- Data Processing Libraries: Pandas and NumPy
- Visualization Library: Matplotlib
- Dataset Format: CSV (Login Attempts / Credential Stuffing Dataset)


Signature of the Guide


Course Instructor